

The rational-spectrum bottleneck in dimension one

An exact later-literature answer

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Status. `literature_already_answered`. This is a literature-identification note, not a claim of a new proof.

Source question

Dutkay and Lai, *Some reductions of the spectral set conjecture to integers*, arXiv:1301.0814v2, page 4, Question (Q1), ask:

Is it true that every spectral set of Lebesgue measure 1 has some rational spectrum?

The same issue is introduced on page 3, immediately after Remark 1.4, as the missing property needed to reverse the implications among the spectral-to-tile statements on $\mathbb{R}$, $\mathbb{Z}$, and all finite cyclic groups $\mathbb{Z}_n$.

Exact supporting answer

Fu and Song, *Rational Spectra for Finite Hadamard Pairs and Bounded Spectral Sets on the real line*, arXiv:2607.19858v1, prove on page 4:

Theorem 1.3. Let $\Omega \subset \mathbb{R}$ be a bounded measurable spectral set with $|\Omega| = 1$, and let Λ be a spectrum of Ω with $0 \in \Lambda$. Then $\Lambda \subset \mathbb{Q}$.

The proof is on page 14. Their Remark 5.1 observes that $0 \in \Lambda$ is only a normalization: for any $\lambda_0 \in \Lambda$, the translate $\Lambda - \lambda_0$ is again a spectrum. Thus every spectral set in the source question has a rational normalized spectrum. This is stronger than the requested existence statement.

The supporting authors explicitly know they are answering the source question. Their introduction cites Dutkay–Lai, describes rationality as the missing hypothesis in that reduction, and states immediately after Theorem 1.3 that the theorem supplies it. The mechanism is periodicity and fiberization to a finite spectral pair, followed by their finite rationality theorem (Theorem 1.2).

Search, prior gap, and scope

The local run indexes had no prior entry for arXiv:1301.0814 or the rational spectrum question. Exact-title, phrase, and arXiv searches located the 2019 preprint arXiv:1908.02794 and the 2026 answer. The 2019 preprint was withdrawn after a gap was found in Section 4, so it is not used as the supporting answer. Fu–Song explicitly discuss that failed claim and replace its critical step with a Galois/modulus-rigidity argument.

This identification closes Question (Q1) only. The full one-dimensional Fuglede conjecture remains open, reduced to its finite cyclic analogues, and Question (Q2) concerning the Coven–Meyerowitz property is not answered here.

References

1. D. E. Dutkay and C.-K. Lai, *Some reductions of the spectral set conjecture to integers*, arXiv:1301.0814v2 (2013).
2. X.-Y. Fu and Z.-J. Song, *Rational Spectra for Finite Hadamard Pairs and Bounded Spectral Sets on the real line*, arXiv:2607.19858v1 (2026).
3. C.-K. Lai and Y. Wang, *Spectrum is rational in dimension one*, arXiv:1908.02794v2 (withdrawn, 2020).